

Rules for logarithms:

$$\log_a xy = \log_a x + \log_a y$$

$$\log_a \frac{x}{y} = \log_a x - \log_a y$$

$$\log_a x^m = m \log_a x$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

1. Simplify the following expressions as far as possible!

(a) $2\sqrt{27} + \sqrt{3}$
 $= 6\sqrt{3} + \sqrt{3} = \underline{\underline{7\sqrt{3}}}$

(b) $\frac{16x^2y^3z}{2xy^2z} = \frac{8xy}{1} = 8xy$

(c) $-3\sqrt{5} + 2\sqrt{20} - 2\sqrt{45} - 2$
 $= -3\sqrt{5} + 4\sqrt{5} - 6\sqrt{5} - 2$
 $= \underline{\underline{-5\sqrt{5} - 2}}$

2. Expand and simplify!

(a) $(3 + 2\sqrt{5})(1 - \sqrt{5})$
 $= 3 - 3\sqrt{5} + 2\sqrt{5} - 10 = \underline{\underline{-7 - \sqrt{5}}}$

(b) $(4 - 3\sqrt{2})(2 - \sqrt{2})$

$$= 8 - 4\sqrt{2} - 6\sqrt{2} + 6$$

$$= \underline{\underline{14 - 10\sqrt{2}}}$$

(c) $x^4(x^2y^2 + 2xy) - 2x^6y^2$

$$= x^6y^2 + 2x^5y - 2x^6y^2$$

$$= \underline{\underline{-x^6y^2 + 2x^5y}}$$

(d) $\frac{1}{x^2}(x+y)^2 - 4(1+x^{-1}y)$

$$= \frac{x^2 + 2xy + y^2}{x^2} - 4\left(1 + \frac{y}{x}\right)$$

$$= \cancel{1} + \frac{2y}{x} + \frac{y^2}{x^2} - \cancel{4} - \frac{4y}{x} = \underline{\underline{-3 - \frac{2y}{x} + \frac{y^2}{x^2}}}$$

3. Solve the following equations.

(a) $\log_4(x) = 2$

$$4^2 = x \Rightarrow \underline{\underline{x = 16}}$$

(b) $\log_{\frac{3}{2}}\left(\frac{27}{8}\right) = x$

$$\left(\frac{3}{2}\right)^x = \frac{27}{8} \Rightarrow \underline{\underline{x = 3}}$$

(c) $2^{3x-1} = \frac{1}{16}$

$$3x-1 = -4 \Rightarrow \underline{\underline{x = -1}}$$

(d) $\log_3(2x+1) - \log_3(3x) + 1 = 2$

$$\log_3\left(\frac{2x+1}{3x}\right) = 1 \Leftrightarrow 3^1 = \frac{2x+1}{3x}$$

$$9x = 2x+1$$

$$\therefore x = \underline{\underline{\frac{1}{7}}}$$

(e) $-\ln(3x-2) + \ln x - 2 = 0$

$$\ln x - \ln(3x-2) = 2$$

$$\ln\left(\frac{x}{3x-2}\right) = 2 \Rightarrow e^2 = \frac{x}{3x-2}$$

$$(3x-2)e^2 = x$$

$$3xe^2 - 2e^2 = x$$

$$3xe^2 - x = 2e^2$$

$$x = \underline{\underline{\frac{2e^2}{3e^2-1}}}$$

4. Write $16^{-\frac{3}{4}}$ as a fraction in simplest form.

e.g. $16^{-\frac{3}{4}} = (2^4)^{-\frac{3}{4}} = 2^{-3} = \frac{1}{2^3} = \underline{\underline{\frac{1}{8}}}$

5. Express with a **positive** integer denominator and simplify as far as possible.

(a) $\frac{2}{\sqrt{7}} = \frac{2\sqrt{7}}{7}$

(b) $\frac{14}{\sqrt{2}-3} \cdot \frac{\sqrt{2}+3}{\sqrt{2}+3} = \frac{14\sqrt{2}+42}{2-9} = \frac{-14\sqrt{2}-42}{7} = \underline{\underline{-2\sqrt{2}-6}}$

$$\begin{aligned}
 \text{(c)} \quad \frac{3+\sqrt{2}}{3\sqrt{2}-1} \cdot \frac{3\sqrt{2}+1}{3\sqrt{2}+1} &= \frac{(3+\sqrt{2})(3\sqrt{2}+1)}{18-1} = \frac{9\sqrt{2}+3+6+\sqrt{2}}{17} \\
 &= \frac{10\sqrt{2}+9}{17}
 \end{aligned}$$

6. The expression below can be written in the form $\frac{a}{7} + \frac{2a}{7}\sqrt{3}$, where $a \in \mathbb{N}$. Find the value of a .

$$\begin{aligned}
 \frac{4+\frac{2}{\sqrt{3}}}{2\sqrt{3}+\frac{1}{\sqrt{3}}} \cdot \frac{\sqrt{3}}{\sqrt{3}} &= \frac{4\sqrt{3}+2}{6+1} = \frac{2+4\sqrt{3}}{7} = \frac{2}{7} + \frac{4}{7}\sqrt{3} \\
 \therefore a &= 2
 \end{aligned}$$

7. Show that $2\log_3 5 - 3\log_3 2 + 1 = \log_3 \left(\frac{75}{8}\right)$.

$$\begin{aligned}
 \text{LHS: } 2\log_3 5 - 3\log_3 2 + 1 &= \log_3(5^2) - \log_3(2^3) + \log_3 3 \\
 &= \log_3\left(\frac{25}{8} \cdot 3\right) = \log_3\left(\frac{75}{8}\right) \quad (\text{shown})
 \end{aligned}$$